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Comparing Presence and Performance Between MR and Screen-Based Flight Simulation Across Varying Cognitive Workload Conditions

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The use of immersive technologies has become increasingly popular within the flight domain, offering benefits such as reduced costs and enabling new measurement methods like eye-tracking. However, it remains unclear exactly how immersive technologies affect performance. In particular, it is uncertain whether the level of cognitive load influences these outcomes, and if so, whether this effect differs from screen-based environments. To investigate this, four scenarios were created based on all combinations of the independent variables' presence and cognitive load. Participants (N=24) completed all conditions, in which they performed three tasks: an n-back task, a reaction time task, and answering questions. Their performance was measured continuously throughout the experiment while presence and cognitive load were assessed after each condition. A significantly higher sense of presence was reported in MR than in the screen condition. A difference was also found in modality for success rate, where participants performed better with screen than with MR. No significant results between modality and screen were found for either performance metric.

SAMMANFATTNING

Användningen av immersiva teknologier har blivit allt mer populär inom flygdomänen och erbjuder fördelar såsom minskade kostnader och möjliggörandet av nya mätmetoder som eye-tracking. Det är dock fortfarande oklart exakt hur immersiva teknologier påverkar prestation. Framför allt är det oklart om nivån av kognitiv belastning påverkar detta och om effekten i så fall skiljer sig mellan immersiva och skärmbaserade miljöer. För att undersöka detta skapades fyra scenarier baserade på alla kombinationer av de oberoende variablerna närvaro och kognitiv belastning. Deltagarna (N=24) testade samtliga scenarier, där de utförde tre uppgifter: ett n-back-test, ett reaktionstest samt besvarade frågor. Deras prestation mättes kontinuerligt under experimentet, medans närvaro och kognitiva belastning skattades efter varje scenario. En signifikant högre känsla av närvaro rapporterades i MR jämfört med skärm. En skillnad observerades också mellan modaliteterna för andel rätt i n-back test, där deltagarna presterade bättre med skärm än med MR. Inget signifikant resultat mellan modalitet och arbetsbelastning observerades för något prestationsmått.

CCS Concepts: • **Human-centered computing** → **Mixed / augmented reality**; **Empirical studies in HCI**.

Keywords: presence, performance, cognitive workload, immersion, immersive technologies, extended reality, mixed reality, flight simulation

Nyckelord: närvaro, prestation, kognitiv arbetsbelastning, inlevelse, immersiva teknologier, utvidgad verklighet, blandad verklighet, flygsimulatorer

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1 INTRODUCTION

Flight simulation has become an established alternative to conventional flight training, offering benefits such as increased safety, allowing dangerous scenarios to be practiced without risk, and reduced costs, as the use of real equipment is minimized [1]. However, with new technologies on the horizon, even more opportunities have arisen. One such development is immersive technologies. With benefits such as lower operational costs, flexibility, efficiency and the possibility to integrate other measurement methods, such as eye tracking and haptics [4, 18, 23], immersive technologies are already becoming increasingly popular within the flight domain [24]. One of the larger benefits of immersive technologies is the increased presence. Given previous research, immersive technologies in general have been shown to contribute to higher levels of perceived presence compared to conventional screen-based systems [10, 22]. Furthermore, prior studies suggest that increased presence can enhance performance and facilitate learning transfer [21, 26], making immersive technologies an interesting alternative for flight simulation training. While this may seem very promising, research also indicates that presence may influence cognitive overload [25, 34] which in turn can negatively affect performance if the demands exceed the user's cognitive capacity[34]. This raises the question of whether the sense of presence can reach a threshold where it turns from beneficial to excessive. Understanding this concept, and how these factors affect each other, is essential for further development since an imbalance can reduce Extended Reality's (XR) advantage over screen-based flight simulators.

Previous research by Parong et al. [22] examined how the level of presence affected spatial learning performance and found that higher levels of presence generally led to better performance. However, the results varied depending on the task, which may have imposed different types of cognitive load. Thus, the study suggests that the level of cognitive load may influence users performance, but it does not investigate the nature of this relationship. Makransky et al. [19] examined participants' learning outcomes in Virtual Reality (VR) compared to desktop environments and found that VR resulted in higher cognitive load and lower learning outcomes. This study suggests a possible relationship between presence and cognitive load, however, it does not examine how different levels of cognitive load may influence learning outcomes. Rather, it only shows that VR was associated with higher cognitive load and lower learning outcomes, without investigating how these differences vary across levels of cognitive load. Sari et al. [25] also examined how cognitive load moderates learning effectiveness in VR and found that lower cognitive load led to better learning outcomes. However, the study did not include a comparison with screen-based environments, nor did it investigate how different levels of cognitive load may influence the advantages that immersive technologies are often suggested to have over traditional screens. Despite these contributions, it remains unclear exactly how cognitive load shapes the effect of presence on performance, and whether this relationship differs between immersive technologies and traditional screen-based environments. In particular, there is a lack of studies directly comparing XR and screen-based conditions in terms of how different levels of cognitive load interact with presence, and whether cognitive load has a stronger impact in XR-environments than in screen-based environments. This gap is particularly relevant as this may undermine the positive effects observed and influence whether immersive technologies can be established as a prominent alternative in the flight simulation domain. Furthermore, most studies have focused on VR [19, 22, 25], yet Mixed Reality (MR) has shown promise as a more effective alternative [33] highlighting the need for further research, particularly in relation to cognitive load and performance.

This study addresses the field of flight simulation and immersive technologies, and specifically addresses the research gap concerning how the use of MR affects performance and whether the level of

cognitive load influences the results. In particular, it investigates whether changes in cognitive load result in measurable differences in task performance between MR and screen-based simulation. The main research question in this study is: *How does cognitive load affect presence and performance in screen-based vs MR flight simulations?* By implementing a 2×2 within-subjects design with cognitive load and presence as the two independent variables, four experimental conditions were created based on all combinations of the variable levels. User performance could then be collected and analysed, making it possible to examine potential interaction effects between the two variables. Findings showed a difference in modality where users felt more presence in MR than with a screen. A difference was also found in modality for success rate, where participants performed better with screen than with MR. No significant difference between modality and screen were found for either performance metric.

2 BACKGROUND

This section explains the concepts of presence, immersion, performance, and cognitive load. It also describes how these factors are connected and influence one another. Furthermore, it explains why these aspects are important in flight simulation training.

2.1 Flight Simulation Training

Flight simulation is today used as an alternative to conventional flight training as a way to both increase safety, since dangerous situations can be practiced without risk, and reduce costs, as real equipment does not need to be used. It also helps decrease environmental impact by saving fuel [1]. However, new technologies have opened up additional possibilities. One such development is immersive technology, whose use is becoming increasingly popular within the flight domain [24]. The use of extended reality (XR) offers several advantages, such as reduced requirements for physical facilities, since physical interfaces like controls and instrument panels can be represented in the virtual environment [4, 23]. In addition, XR has lower operational costs than dome-based simulators, making it a more cost-effective option [4]. Furthermore, it is more flexible by allowing the functionality and design to be dynamically adapted [4, 18]. It also enables the integration of new measurement methods, such as eye tracking and haptics [4]. Studies have shown that the use of VR is a promising alternative for flight training [12, 16, 32]. For example, one study by Lawrynczyk comparing virtual reality flight training with screen-based flight training, compared user experience, mental workload, and flying performance metrics and found VR to show promising results as an additional tool [16]. In another study, Trinon conducted both a market research as well as a competitive analysis and also found it to be a promising tool for pilot training [32]. A third study by Dalladaku et al. found no significant drop in performance between virtual reality flight training and traditional flight simulator training and although further research is needed, the results suggest that VR is a good alternative worth investigating further [12]. To summarize, studies are being conducted to examine the potential impact of XR within industry, and while the general trend is promising, more research is still needed to fully establish immersive technologies as a viable alternative [24].

Another reason why the use of immersive technologies might be a good alternative to more traditional flight simulation is that studies have shown that a high sense of presence can support positive learning transfer. When a virtual environment successfully creates both the feeling of being there and the illusion that the scenario is actually occurring, users are more likely to behave as they would in the real world

[27], thus enabeling better learning transfer. It is not only important that the simulation looks real, it also needs to function like the real task, so users act the same way as they would in reality [14].

Together, these findings make presence an important aspect to consider when designing for flight simulation training and therefore it is important to examine how it differs between traditional flight simulators and XR-based flight simulators.

2.2 The importance of presence

The concept of presence has been defined as a person's subjective experience of being in an environment, and when talking about presence in XR, it refers to a person's perception of being physically present in the virtual environment rather than in the physical world [35]. A higher presence means a greater sense of being in the virtual world.

Presence is thought to be one of the key features of virtual environments. Previous studies have shown that a higher perceived presence is associated with improved learning outcomes when working in an educational virtual environment (EVE) [21, 26]. For example, in a systematic review of seventeen studies, it was found that higher levels of presence correlated with the achievement of learning objectives [26]. A high sense of presence is thought to increase the similarity between users' responses in virtual environments and their responses in real-life contexts, thus improving the effectiveness of mediated environments for purposes such as learning and training [14, 27]. Earlier research conducted by Slater et al. examined the factors that contribute to the experience of presence in virtual environments, suggesting that users tend to feel more present when they can move naturally, when the environment responds consistently to their actions, and when perspective and movement feel realistic [28].

In other words, higher levels of presence are associated with more realistic user behavior, such that users tend to respond to virtual events in ways similar to real-world situations. Together, these findings support the view that presence is a key factor in enabling virtual environments to function effectively as tools for training and learning.

2.2.1 The impact of Immersion. The concept of immersion has been defined in many ways both as a subjective and objective measurement. In some cases, immersion is used interchangeably with presence, describing the user's subjective experience of presence within a virtual environment, for example by McGloin et al. [20]. Immersion has also been described as an objective measure that refers to how immersive the technology used to create the virtual environment is [29]. This definition was used in a study by Cummings and Bailenson who examined the relationship between immersion and presence. The study conducted a meta-analysis of previous research to examine how immersive technologies influence users' sense of presence. It aimed to determine whether higher levels of immersion in mediated environments led to stronger feelings of presence and thereby enhancing the overall user experience. Results showed that higher immersion, that is, the system is better able to present a convincing virtual environment, in turn contributes to a greater sense of presence, all in all indicating that the level of immersion a system can provide affects the users' perceived presence [10].

Given this research, it can be assumed that immersive technologies, i.e., extended reality, may contribute to a higher level of perceived presence compared to conventional screen-based technologies. In turn, this raises the question of whether the use of XR can lead to improved training conditions and, in turn, enhanced performance. Therefore, it is of interest to conduct further research on how users' perceived presence is affected in XR compared to screen-based solutions.

2.2.2 The relationship between immersion, presence and performance. To explore the relationship between immersion, presence, and performance, Parong et al. investigated how the experience of presence influences the effects of immersive technologies on learning and task performance. The users got to complete a set of tasks in three different levels of immersive technology and afterwards filled out a questionnaire to measure their perceived presence. Results showed that a higher immersion condition led to better performance as well as higher levels of presence. It also showed that a higher level of presence generally led to better performance. However, the effect varied depending on the type of task, suggesting that the link between immersive technology and learning is not straightforward and may be influenced by perceived presence and cognitive load [22].

Another study by Chandio et al. aimed to investigate the relationship between presence and reaction time in mixed reality (MR) as a way to examine whether reaction time could serve as an objective measure of presence, given that the questionnaires traditionally used are considered to have significant limitations. By manipulating the environment, the researchers were able to influence users' sense of presence, and by measuring their reaction time in response to visual stimuli, they examined the relationship between presence and reaction time. The study found that the higher the level of presence experienced by the user, the faster their reaction time, suggesting that reaction time could be used as an objective measurement [7]. This in turn suggests that presence may be related to performance.

Building on their study, Chandio et al. further investigated how distractions influenced their findings. In an additional study, they examined both coherent distractions, in the form of distractions integrated into the virtual environment, and conflicting distractions, in the form of distractions originating from the real world. The results showed that both types of distractions led to increased reaction times, with a larger increase observed for conflicting distractions, suggesting that presence acts as a mediating factor. In addition, cognitive load increased in both conditions. Overall, the study supports the idea that reaction time and presence are closely related, such that lower levels of presence are associated with longer reaction times. The results also indicate that presence plays a role in how cognitive load affects reaction time, supporting the idea that cognitive load is related to presence. It showed that an increase in cognitive load, led to less presence and thus slower reaction time, i.e. worse performance [8].

These findings highlight the close relationship between cognitive load and presence in immersive technologies. While XR can enhance presence, excessive cognitive load may reduce it, potentially affecting performance. Understanding how these factors interact is especially important in complex flight situations, as an imbalance could undermine the advantages of XR compared to conventional screen-based simulators.

2.3 Cognitive Load

The Cognitive load theory (CLT) was developed by John Sweller and refers to the limited capacity of working memory during learning [9, 30, 31]. There are three types of cognitive load; intrinsic, extraneous, and germane. Intrinsic load refers to the inherent difficulty of the task, extraneous load refers to unnecessary distractions that hinder learning, and germane load refers to the mental effort dedicated to actually understanding the information [9].

Studies have shown that designing for high presence does not only have positive effects. An increased level of presence can also create a cognitive overload for the user [25, 34]. One study by Whitelock et al. showed that a higher level of presence can make the learning environment more engaging and easier to use, but it can also make it harder to understand the content due to cognitive overload, as too much of the

user's attention is directed toward other stimuli and the experience itself, rather than learning [34]. By giving the user too much information at once, it can overload one's working memory capacity, leading to worse performance and increased error rates. One way to experimentally increase cognitive load is therefore to exceed working memory capacity by requiring users to handle multiple things at once, or by increasing how frequently tasks appear.

VR-based simulations often provide a greater sense of presence, however it is still not entirely clear how the level of presence in VR specifically influences learning transfer in relation to cognitive load. The use of VR has in some studies been associated with higher cognitive load and lower learning outcomes [19, 25]. For example, Makransky et al. found that although students felt more present in the VR condition compared to a desktop condition, they showed lower learning outcomes and experienced significantly higher cognitive load [19]. In addition, Lawrynczyk compared VR flight training to screen-based simulation and found that several measures of cognitive load were significantly higher in the VR condition [16].

These findings make it interesting to further investigate how cognitive load influences perceived presence in XR compared to screen-based flight simulation, and how this, in turn, affects user performance in the two technologies.

2.4 Mixed Reality vs Virtual Reality

While much research has focused on VR, several studies have begun to explore the differences between VR and MR, providing insight into how each technology affects presence and user experience. For instance, Dahlkvist investigated the use of XR technology in flight simulation and compared VR and MR in terms of usability, immersion, presence, and simulator sickness. The results showed that MR was perceived as more immersive and had higher usability when interaction was involved. Meanwhile, VR was found to be more immersive in scenarios where no interaction was required. However, it is important to note that the study did not include significance testing [11].

Furthermore, Wang compared VR and MR in flight simulation training in order to investigate which technology provides better psychological and ergonomic fidelity. Participants completed a task that involved pressing buttons in a virtual interface, and the number of correct and incorrect presses was recorded to assess ergonomic fidelity. Afterwards, participants completed a questionnaire about their experience based on the PQ Questionnaire [35] to assess psychological fidelity. The study found that performance was higher in MR, and users also preferred MR when considering aspects like adaptation, movement, and concentration. The study suggests that MR provides higher psychological and ergonomic fidelity than VR and is thus better suited for flight training [33].

In another study by Lindberg, results showed that MR can result in lower levels of cybersickness [17], a phenomenon that can occur within XR when there is a mismatch in sensory information. Cybersickness refers to the negative side effects users can experience after interacting with a simulated environment, such as nausea or eye strain, and is often explained by the sensory conflict theory [13].

Given the focus on VR in previous research, and the advantages observed within MR, it is natural to extend the comparison to mixed reality to better understand how different XR technologies could serve as an alternative to screen-based flight simulation. Specifically, how do presence and performance differ between screen-based and MR systems, and what role does cognitive load play in shaping these outcomes? Understanding these relationships could provide important insights into the potential advantages and limitations of MR in the aviation domain.

3 EVALUATION

To investigate how user performance differs between MR and screen-based flight simulation, and whether these differences vary depending on the level of cognitive load imposed on the user, a 2×2 within-subjects experimental design was used. The experiment was designed to address the following research question: *How does cognitive load affect presence and performance in screen-based vs MR flight simulations?*

3.1 Participants

A total of 24 company employees (15 male, 9 female), aged 23 to 65 (M = 36, SD = 14.6), were recruited through convenience sampling. Regarding prior XR-headset experience, 4 participants reported having none, 10 little, 6 some, and 4 reported extensive experience. Those with prior experience stated it was primarily due to professional use, participation in previous studies, or personal gaming. A similar distribution was observed for flight simulator experience where 4 participants reported having no experience, 7 little, 6 some, and 7 extensive. Sources of experience were similar, with reasons such as professional use, prior study participation, or personal gaming. Since all participants were recruited from within the company where employees generally know quite a lot about the subject, the generalizability of the results may be limited. Furthermore, no compensation was offered for participation other than the incentive of being able to try a flight simulator and a XR headset.

3.2 Study Design

This study was designed to investigate how presence affects performance, and further, how the level of cognitive load influences these outcomes. To allow for direct comparisons between conditions and minimize the influence of individual differences, the experiment followed a 2×2 within-subject design with cognitive load and presence as the two independent variables, resulting in four experimental conditions based on all combinations of the variable levels. These conditions consisted of two levels of cognitive load (high and low) and two modalities (MR and screen). All participants were exposed to each condition for five minutes through a flight scenario in which the participant completed various tasks as described in section 3.2.1. Data were collected both from participants' task performance and through standardized questionnaires administered after each condition. The order in which participants experienced the conditions was determined using a balanced Latin square design in order to reduce potential learning effects and fatigue that could arise during the course of the experiment.

Parong et al. found that higher immersion led to higher levels of presence as well as better performance [22]. On the other hand, Makransky et al. found that while students felt more present in VR than on a desktop, they had significantly higher cognitive load in the VR condition and lower learning outcomes [19]. Thus this study is designed to investigate the following hypothesis:

(1) H1: Presence will be higher in both low and high cognitive load environments for MR-based simulation when compared to screen-based simulation.
(2) H2: Participants will perform better in MR compared to screen-based simulation when cognitive load is lower, in terms of performance.
(3) H3: There will be a smaller difference in performance, between MR vs Screen-based simulation, when the cognitive load is high compared to when the cognitive load is low.

The performance is determined by means of lower error rate, higher success rate, quicker reaction times and higher response rate.

3.2.1 Task Design. The study used two different levels of cognitive load where one level is designed to produce lower cognitive load and the other to produce higher cognitive load. Both levels were tested with both the MR set up and the screen-based set up. This way, data is obtained on both the users perceived presence with screen vs MR, how the users performed, and whether, and if so how, these results vary with changes in cognitive load.

To increase cognitive load, the frequency of one task was increased as well as the difficulty of another, reflecting a realistic scenario in which pilots must perform multiple tasks in quick order while maintaining high situational awareness. Table 1 presents the design of the two levels.

Table 1: Task Design for the two levels

Level 1: Low Cognitive Load	Level 2: High Cognitive Load
2-back test	3-back test
Reaction time test in 10 seconds intervals	Reaction time test in 10 seconds intervals
6 questions to answer during the flight	10 questions to answer during the flight

Description of the tasks:

- **n-back test:** In an n-back task, participants are shown a continuous sequence of stimuli and must respond when the current stimulus matches the one n steps earlier in the sequence. Throughout the level, participants flew through a series of objects on either side of them, each displaying a number. The numbers appeared when an object entered the participant's proximity and remained visible for three seconds. At any given time, only one number was presented, always on the object closest to the participant. The participants' task was to memorize the sequence of numbers and respond when the currently displayed number matched the one presented two steps earlier (2-back condition, representing lower cognitive workload) or three steps earlier (3-back condition, representing higher cognitive workload). Responses were recorded via a button press on the joystick. This task was designed to place a load on the user's memory.
- **Reaction time test:** During the game, participants heard an auditory cue at regular intervals. Upon hearing the sound, they were required to press a button on the joystick as quickly as possible. This task was designed to capture their attention, measure reaction time, and require them to respond rapidly while simultaneously performing other tasks, thereby simulating divided attention and multitasking.
- **Question responses:** The participants were asked questions through a prerecorded recording in either the headset (for MR) or the speakers (for screen) during the flight. The questions required the users to be aware of their surroundings and were created to further increase the cognitive workload, both by requiring participants to have situational awareness and as a way to create a distraction from the other tasks. All questions focused on the plane's current status such as fuel and altitude, and in order to answer, the participants had to look at the display in front of them to obtain the right answer and say it out loud.

By mixing several tasks and varying their frequency, the study and the different conditions were designed according to the requirements, i.e. high or low cognitive load. It is important to note that no real pilots participated in the study, which limited the ability to create fully authentic scenarios, since the tasks had to be designed in a way that did not require prior experience. The tasks were therefore developed to both resemble realistic activities as closely as possible and to create a lifelike scenario

for a pilot to the extent feasible, requiring memory recollection, maintaining situational awareness, and multitasking.

3.2.2 Design Considerations. A delimitation of the study was that the scenarios were designed with the aircraft put in autopilot. This was done for several reasons. First, it minimized the risk of learning effects across sessions. Second, it ensured users stayed on the intended path. If a user were to accidentally steer away from it, they would miss parts of the course and data would be lost. Third, it was important to avoid simulator sickness as much as possible so that the results were not affected by users feeling unwell or needing to interrupt the experiment early. For the same reason, each flight session was limited to five minutes. The tasks place a high cognitive load on the user, and there is a risk that fatigue increases over time. To minimize potential effects of learning and fatigue in the final flight session, the duration was kept as short as possible while still trying to give enough time for participants to develop a sense of presence. A pilot study was conducted to determine an appropriate duration for achieving this balance. While this delimitation might decrease the realism in the scenario, it was deemed necessary in order to maintain experimental control and ensure reliable data collection.

3.2.3 Story Design. To maximize the chances of developing presence and the feeling that the participants are flying in a real aircraft, even though the plane can't be maneuvered, a story surrounding the experiment was created. The users were told that the plane is put on autopilot and that they have a lot of other tasks to focus on in order to succeed in the mission. Since the experiment is not conducted with real pilots, putting the plane in autopilot still functions as a credible and realistic scenario. The participant receives both visual input through the environment and the cockpit they're in, as well as auditory input from the background noise and radio calls through the headphones. Together this was meant to create a feeling of actually being in a plane.

All together these elements aimed to contribute to making the scenario feel more realistic as the users need to be aware of their surroundings at all times in the same way that real pilots have to even though the plane itself is not maneuvered. The limitations that were made as well as the components added are valuable and necessary in order to create the best possible environment for the experiment.

3.3 Measures

To compare the different conditions both subjective and objective measurements were collected. The n-back task was divided into two performance metrics, error rate and success rate. During the experiment, the following dependent variables were measured:

- **Error Rate:** Number of incorrect responses divided by the number of non-targets in the n-back test.
- **Success Rate:** Number of correct responses divided by the number of targets in the n-back test.
- **Reaction Time (ms):** Reaction time was defined as the time elapsed between the onset of the auditory cue and the participant's button press. Each response was recorded, and the mean reaction time for each participant was calculated and used in the analysis.
- **Response Rate:** The percentage of questions correctly answered during the flight.
- **Perceived Cognitive Load:** After each condition, the participants got to assess their perceived workload. This was done with the NASA Task Load Index [3] and included scales for mental demand, physical demand, temporal demand, performance, effort and frustration level. Because the ratings were unweighted, the Raw Task Load Index (RTXL) was applied in the analysis.

- **Perceived Presence:** After each condition, the participants also got to assess their perceived presence, the subjective feeling of 'being there', on a scale from -3 (lowest) to 3 (highest). This was done using the Igroup Presence Questionnaire (IPQ) [2]. The IPQ questionnaire was divided into its subscales, general presence, spatial presence, involvement, and experienced realism, which were then calculated according to the official source [2].
- **Simulator Sickness Questionnaire:** Simulator sickness was assessed after each condition using the SSQ questionnaire [15]. The SSQ scores were separated into subsections and calculated according to Bimberg et al. [5]. A total SSQ score between 20-30 is considered minimal to moderate motion sickness, while scores above 40 indicate "a bad simulator" [6]. This measure was collected as a control variable to determine whether the results may have been influenced by participants' experience of simulator sickness.

After each condition, participants were asked to answer a few questions about their experience and subjective impressions of the conditions and after all conditions were completed, they were asked additional questions about their overall experience. All performance metrics were automatically recorded by the computer, except for response rate, which were collected manually. All subjective questionnaires were administered and completed digitally.

3.4 Apparatus

The following section describes the technical implementation of the study and the technology used.

3.4.1 Technical Implementation. The conditions were built upon already existing material in Unreal Engine and components were added in order to create a feasible scenario for the study.

n-back test. To present participants with a continuous sequence of stimuli, in this case numbers, objects on which the numbers would appear were placed in the environment at approximately equal distances from one another. With slight variation, to enhance realism and create a more dynamic experience. They were also distributed across different positions along the x and y-axes. A blue panel was positioned on the side of each object to clearly indicate where the number would appear, see picture 1. The numbers were programmed to appear when the object was relatively close to the user, which was determined based on predefined timing established through pilot testing. The number sequences used in the n-back task differed across conditions and were generated through randomization. However, manual adjustments were made to ensure an equal number of target stimuli across conditions. To maintain comparable levels of difficulty between screen and mixed reality in the same cognitive load condition, the timing of target occurrences followed the same pattern in these conditions, although the specific numbers differed. To prevent participants from recognizing patterns, the target sequences were not identical between high and low load conditions, but only within each load level.

Reaction time test. An auditory cue was implemented and played at 10-second intervals. Constraints were applied to ensure that only one response was recorded per sound event. This was done to prevent invalid data from being registered in cases where participants pressed the button multiple times or pressed an incorrect button.

Response test. Using ElevenLabs and Audacity, a recording was created to present participants with questions at regular intervals. In the low load conditions, questions were presented approximately every 50 seconds, resulting in six questions per flight. In the high load conditions, questions were presented

approximately every 30 seconds, resulting in ten questions per flight. Questions included level of altitude the plane was flying at, fuel, whether the participant is currently flying over water or land, how many and which missiles are in possession as well as how many entities, enemies and friendly, the user can see in their radar. In order to be able to ask the last question, entities were programmed into the radar beforehand.



Figure 1: The participants point of view during the experiment

3.4.2 Technical Set-up. The computer used in the study had a NVIDIA GeForce RTX 4090 and an i9 Intelprocessor. For the screen-based conditions, the size of the screen used were 49” with a 1000R curvature and the sound was played through two stereo speakers placed on the floor in front of the participant. In MR conditions a Varjo XR-4 was used and sound were played through the Varjo headset. The head down display had a size of 27”. See figure 2.

3.5 Procedure

The experiment was carried out at OpenFLSC and each session lasted approximately 90 minutes. At the beginning of the session, all participants were asked to fill out a consent form, which provided information about the study, their participation, and the handling of their data in accordance with GDPR guidelines. They were then given a brief introduction to the structure of the experiment and any questions that arose were answered. Participants were not informed in advance about the order in which they would complete the different conditions. Before the experiment began, participants were given a tutorial which included an explanation of where the numbers would appear in the n-back task, where to find the responses to the questions they would be asked, and which buttons to use for both the n-back and reaction time tasks. Participants then completed a two-minute practice session to ensure that they understood the tasks and felt prepared to begin.

When the participant felt ready the first session began. Each participant tested all of the four different conditions for five minutes each and after each session, they filled out the questionnaires mentioned in section 3.3. Before each session with MR, the users got to try on the headset and adjustments were made to ensure a comfortable fit on the participant’s head. The system was also calibrated to account for participants’ eye positions.



(a) Set-up with screen **(b) Set-up with MR**

Figure 2: Pictures showing the two different set-ups

After all conditions were tested, the participants got to fill out an additional form with open ended questions about their overall experience. Furthermore, basic demographic information was collected as well as information regarding their previous experience with flight simulators and XR-headsets. Afterwards the participants were thanked for being part of the study and were given the opportunity to ask questions and talk about the experiment.

3.6 Data Analysis

Before analysis, a Shapiro-Wilk test was conducted on each condition to assess the assumption of normality, and Mauchly's test of sphericity was used to verify that the sphericity assumption was met. Where sphericity was violated, Greenhouse-Geisser corrected p-values were used. The obtained data were then analysed using a 2×2 repeated measures ANOVA with visual modality (MR vs Screen) and cognitive load (low vs high) as within-subject factors. Significant effects were followed up with Bonferroni-corrected paired-samples t-tests ($\alpha = .025$) to examine the modality difference at each load level separately. Where the assumption of normality was violated in one or more conditions, an Aligned Rank Transform ANOVA (ART ANOVA) was conducted as a non-parametric alternative, with follow-up Wilcoxon signed-rank tests with Bonferroni adjustment ($\alpha = .025$). All analysis was done using Python with packages pandas, SciPy, Pingouin, NumPy and Matplotlib. Open-ended responses were analyzed descriptively, with recurring themes summarized to illustrate key findings.

4 RESULTS

Results showed that MR produced a significantly higher sense of presence than the screen condition, and that the high load level was significantly higher than the low load level. A difference was also found in

modality for success rate, where participants performed better with screen than with MR. Furthermore, no significant results between modality and cognitive load were found for any performance metric. All results are outlined in detail below.

Table 2: Table showing the median and interquartile range (IQR) values for all performance metrics.

	Error Rate (%)		**Success Rate (&)**		**Reaction Time (ms)**		**Response Rate (%)**	
	Md	IQR	Md	IQR	Md	IQR	Md	IQR
MR Low	3.4	6.9	80	20	860	230	92	17
Screen Low	3.4	4.3	85	20	920	250	100	17
MR High	3.4	6.9	55	33	1000	330	90	10
Screen High	3.4	3.5	60	13	970	200	90	10

Table 3: An overview of the SSQ scores with subsections showing median and interquartile range (IQR) values. A total SSQ scores between 20-30 is considered minimal to moderate motion sickness, while scores above 40 indicate "a bad simulator" [6].

	SSQ							
	Total		Nausea		Oculomotor		Disorientation	
	Md	IQR	Md	IQR	Md	IQR	Md	IQR
MR Low	24.3	40.2	9.5	31.0	19.0	39.8	27.8	59.2
Screen Low	11.2	15.9	9.5	19.1	7.6	22.7	0.0	13.9
MR High	29.9	69.2	28.6	47.7	26.5	55	27.8	94.0
Screen High	7.5	24.3	9.5	28.6	3.8	24.6	0.0	13.9

Table 4: An overview of the IPQ scores with subsections, GP = General Presence, SP = Spatial Presence, INV = Involvement, ER = Experienced Realism, where the scale ranged from -3 (lowest) and 3 (highest). As well as mean and standard deviation for NASA-RTLX scores.

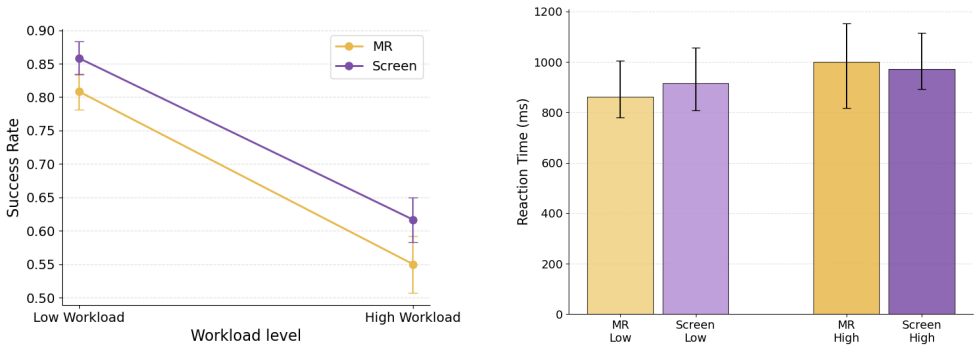
	IPQ								**Nasa-RTLX**	
	GP		SP		INV		ER		Total	
	Md	IQR	Md	IQR	Md	IQR	Md	IQR	M	SD
MR Low	2.0	1.0	1.5	1.0	1.4	1.4	-0.5	1.1	47.0	13.0
Screen Low	-1.5	2.3	-0.7	1.5	0.0	1.3	-1.5	1.1	44.9	12.9
MR High	1.5	1.0	1.6	1.2	1.4	2.1	-0.5	0.8	57.1	11.9
Screen High	-2.0	2.3	-1.2	1.9	-0.4	1.5	-1.4	1.1	54.5	9.5

4.1 Quantitative Data

4.1.1 Error Rate (%). The interaction between modality and cognitive load were significant ($F(1, 23) = 6.4$, $p = .019$). However, follow-up Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .025$) showed no significance at either MR Low vs Screen Low ($p = .878$) or MR High vs Screen High ($p = 1.0$). No difference in either main effect, modality ($F(1, 23) = .92$, $p = .35$) or cognitive load ($F(1, 23) = 3.4$, $p = .076$), were found. Results showed that the error rate between MR (Md = 3.4, IQR = 6.9) and screen (Md = 3.4, IQR = 4.3) were the same, as were the error rate between low load (Md = 3.4, IQR = 6.9) and high load (Md = 3.4, IQR = 3.5). Looking at the conditions, users performed similarly in every condition with MR Low (Md = 3.4, IQR = 6.9), Screen Low (Md = 3.4, IQR = 4.3), MR High (Md = 3.4, IQR = 6.9) and Screen High (Md = 3.4, IQR = 3.5).

4.1.2 Success Rate (%). The findings showed a significance in both main effects, modality (F(1, 23) = 8.9, p = 0.0067) and cognitive load (F(1, 23) = 73.4, p < .001), where users performed better with screen (Md = 80, IQR = 30) than MR (Md = 70, IQR = 40) and better under low load (Md = 80, IQR = 10) than high load (Md = 60, IQR = 30). No significant interaction effect between modality and cognitive load were found (F(1, 23) = 3.7, p = 0.0681). Users performed better with a screen than with MR in both load conditions, and the difference in median value between MR and screen were the same in both conditions, as can be seen in table 2. Follow-up Wilcoxon signed-rank tests were not conducted as the two significant effects contained only two levels.

4.1.3 Reaction Time (ms). The results showed a significance in cognitive load (F(1, 23) = 10.7 , p = 0.0034), where users performed better in low load (Md = 900, IQR = 250) than high load (Md = 980, IQR = 270). No differences were found in modality (F(1, 23) = .38 , p = 0.55) where the median reaction time was similar for MR (Md = 940, IQR = 250) and screen (Md = 960, IQR = 270). Furthermore, no significant interaction effect between modality and cognitive load were found (F(1, 23) = .069 , p =.79).



(a) Success rate per condition

(b) Reaction time per condition

Figure 3: Success rate and reaction time across modality and cognitive load conditions

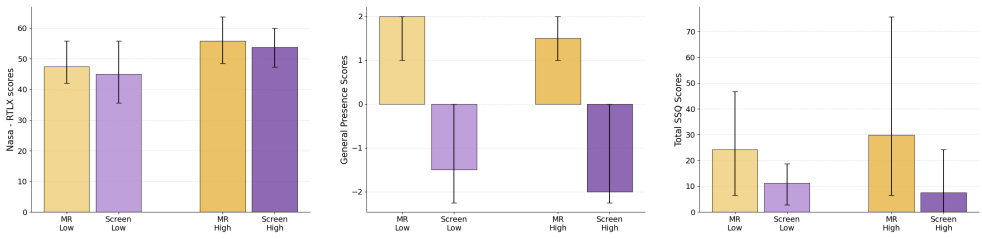
4.1.4 Response Rate (%). Results showed no difference in either main effect, modality (F(1, 23) = .17 , p = .68) or cognitive load (F(1, 23) = 0.016 , p = .90). Looking at individual conditions, performance was similar between MR Low and Screen Low, while identical between MR High and Screen High as can be seen in table 2. No significant interaction effect between modality and screen were found (F(1, 23) = 3.94 , p = .059), thus no follow-up Wilcoxon signed-rank tests were conducted.

4.1.5 NASA-RTLX: Perceived Cognitive Load. A significance in cognitive load was found (F(1, 23) = 35.26, p < .001, ηg^2 = .15), where the low load condition produced significantly lower scores than the high load condition. No significant effect were found in modality (F(1, 23) = 1.86, p = .19, ηg^2 = .01), indicating that presence alone did not influence the scores. Users reported the lowest value for low load (M = 45.9, SD = 12.8) and the highest value for high load (M = 55.8, SD = 10.8), while the difference between MR (M = 52.1, SD = 13.3) and screen (M = 49.7, SD = 12.2) were more similar. The interaction between modality and cognitive load was not significant (F(1, 23) = 0.039, p = .84, ηg^2 < .001), suggesting that the difference between MR and Screen did not vary across the different load levels. Looking at individual conditions, users reported lower cognitive load for Screen Low (M = 44.9, SD = 12.9) and MR Low (M = 47.0, SD = 13.0), and higher cognitive load for Screen High (M = 54.5, SD

= 9.5) and MR High (M = 57.1, SD = 11.9), reflecting the strong effect of cognitive load level across both modalities. Post-hoc comparisons were not conducted as the only significant effect (cognitive load) contained only two levels.

4.1.6 IPQ: Perceived Presence. Results showed that a difference was found in modality ($F(1, 23) = 47.9$, $p < .001$) where users felt more presence in MR (Md = 2.0, IQR = 1.0), than screen (Md = -2.0, IQR = 2.3), regardless of load level. Furthermore, the interaction effect between modality and cognitive load were significant ($F(1, 23) = 12.3$, $p = .0019$) and follow-up Wilcoxon signed-rank test with Bonferroni correction ($\alpha = .025$) showed a significance in both MR Low vs Screen Low ($p < .001$), as well as between MR High vs Screen High ($p < .001$). No difference was found in cognitive load ($F(1, 23) = 0.69$, $p = .42$), suggesting that the level of cognitive load on its own did not affect users' sense of presence. Users reported similar scores for both low load (Md = 0.5, IQR = 4.0) and high load (Md = 1.0, IQR = 4.0).

4.1.7 SSQ: Simulator Sickness. Results showed a difference in modality ($F(1, 23) = 13.7$, $p = 0.0012$) , with users reporting a higher simulator sickness in MR (Md = 26.2, IQR = 53.3) than with screen (Md = 7.48, IQR = 19.6). A significant interaction effect between modality and cognitive load ($F(1, 23) = 9.8$, $p = .0047$) were also found. Follow-up Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .025$) showed a significant difference in both MR Low vs Screen Low ($p = .025$) and MR High vs Screen High ($p = .0012$) showing that there is a significance on both levels, although stronger in the higher level. This is further supported by the median values of the individual conditions. Users reported lower simulator sickness for Screen Low and Screen High, and higher levels of simulator sickness for MR Low and MR High, as can be seen in table 3. No difference was found for cognitive load ($F(1, 23) = 2.2$, $p = 0.15$), with comparable scores between low load (Md = 14.96, IQR = 28.1) and high load (Md = 14.96, IQR = 44.9).



(a) Nasa - RTLX scores **(b) General presence score (IPQ)** **(c) Total SSQ scores**

Figure 4: Plots showing cognitive load, presence and simulator sickness

4.2 Qualitative Data

4.2.1 Perceived usability. 9 people reported that they liked using the headset since it helped keep them focused on the tasks, while 6 people reported that they saw the display better without the headset with 1 person saying that the graphics sometimes felt a bit off in MR. A few people mentioned that they didn't like that they didn't get to steer the plane themselves, while others mentioned that as a positive aspect. Many people (11-14 on each condition) mentioned that the hardest part was to remember numbers for the n-back tasks while simultaneously getting asked questions and that answering questions made them lose track of the number sequence. Some people also mentioned that this made them distracted

and resulted in break in presence. Overall, people reported having a higher presence and more focus with the headset while the biggest risk for loss of focus or break in presence were a mess up in task.

Most participants (15) reported MR Low as their preferred condition, whilst it was very similar between MR High (10) and Screen High (12) for the least preferred condition. The biggest reason for marking a condition as the least favorite was due to it being the hardest.

5 DISCUSSION

The findings support **H1:** *Presence will be higher in both low and high cognitive load environments for MR-based simulation when compared to screen-based simulation.* The findings do not support **H2:** *Participants will perform better in MR compared to screen-based simulation when cognitive load is lower, in terms of performance* or **H3:** *There will be a smaller difference in performance, between MR vs Screen-based simulation, when the cognitive load is high compared to when the cognitive load is low*. Although possible trends are identified aligning with hypothesis H3. This chapter further discusses the results.

5.1 Performance in MR vs Screen

Selverian and Hwang conducted a systematic review and found that higher levels of presence correlated with the achievement of learning objectives [26], and Chandio et al. measured people's reaction time in different MR environments and found that users performed better when experiencing a higher presence [7]. Suggesting that a higher perceived presence could correlate to higher performance. Therefore it can be argued that participants should perform better with an MR simulator than with a screen-based simulator, however, this was not the case in this study. Success rate showed results contradicting hypothesis **H2:** *Participants will perform better in MR compared to screen-based simulation when cognitive load is lower, in terms of performance*, with screen providing better results than MR. In addition, looking att response rate, a pattern is observed also indicating that users might perform better with the screen as can be seen in table 2. Reaction time on the other hand showed a pattern aligning with hypothesis H2 with MR possibly producing quicker reaction times than screen when the cognitive load is lower, see table 2. Error Rate gave the same median value for both MR Low and Screen low, and whilst a significant interaction effect was found, there was no significant difference between MR Low and Screen Low. Furthermore, the SSQ scores showed a significant effect of modality ($p = 0.0012$) with MR providing higher simulator sickness than screen. Although, it is important to note that the score still ranks very low as can be seen in table 3, and together with the low scores for screen, this suggests that the results should be interpreted carefully and that it is unlikely that simulator sickness meaningfully influenced task performance.

Due to the mixed results, and no difference between MR Low and Screen Low in either performance metric, except success rate, H2 is not supported. An interesting observation is that it was specifically the reaction time performance metric that resulted in a pattern indicating that users might perform better in the MR Low environment, which is the same performance metric used by Chandio et al. [7]. This raises the question of whether there may also be a link between the type of task performed in the virtual environment and performance, rather than performance only being influenced by perceived presence or cognitive load. Parong et al. found that the effect of presence on performance varied depending on the type of task [22], which is also indicated by the findings in this study.

5.2 The effect of presence on performance

Cummings and Bailenson found that a higher immersion led to a greater sense of presence [10], indicating that MR should provide a higher presence than screen, which in this study turned out to be true. Users reported feeling significantly more present in the MR environment ($p < .001$) which is also reflected in the general presence score for each condition, which can be seen in table 4. This further supports hypothesis **H1:** *Presence will be higher in both low and high cognitive load environments for MR-based simulation when compared to screen-based simulation.* However, as discussed in section 5.1, the use of MR did not result in better performance than with a screen, contradicting previous findings regarding the positive relationship between presence and performance [7]. The results in this study found that only in the performance metric success rate were there a difference between modalities, where screen performed better. One reason for this might be due to technical difficulties. A few people reported seeing the display better with the screen than the headset which might have affected the results. Furthermore, if there is a mismatch in the masking between the real world and the virtual world, this might cause distractions. Only two people reported this, but it is still worth mentioning. On the other hand, a few people also mentioned they saw the display very well with the headset and some people reported feeling more focused in the virtual world as it shut out other distractions. Therefore no conclusion can be drawn regarding if any technical difficulties might have affected the results, but it is a possibility. Another reason for the results might be due to the limited interaction in the virtual world. Since the participants were not allowed to interact with anything other than the joystick, this might have hindered the positive effects that immersive technologies are shown to provide. As noted by Slater, in order to act in a virtual environment as you would in the real world, it is important to both have a feeling of 'being there', and that the events feel as though they are actually occurring [27]. Limiting interaction to a joystick may have meant that these conditions were not fully met, which in turn possibly reduced the positive effects of the immersive environment. Even though users reported a higher sense of presence in the MR environment, these results might indicate that not only is it enough that you feel more present in the environment, perhaps the positive effect of presence on learning transfer and performance lies in the realism of the interactions. The feeling of actually operating in the world, and not only being there. In other words, presence is important to make people act in the virtual world as they would in the real world, thus creating a scenario where people learn and therefore increasing the potential of learning transfer. However, since participants were not operating in this world, that part was lost. So even though the goal of creating a higher sense of being there was met, this scenario did not enable that type of learning.

5.3 The difference in performance between load levels

Whitelock et al. found that a higher level of presence can lead to a cognitive overload [34], which is also in line with the findings of Lawrynczyk, who compared VR flight training to screen-based simulation and found that several measures of cognitive load were significantly higher in the VR condition [16]. Furthermore, Makransky et al. found that in a VR environment, producing higher presence compared to a desktop condition, participants both experienced higher cognitive load but also showed lower learning outcomes [19]. It can be argued that this may, in turn, reduce the positive effect of presence on performance, which forms the basis of hypothesis **H3:** *There will be a smaller difference in performance, between MR vs Screen-based simulation, when the cognitive load is high compared to when the cognitive load is low.* The findings in this study suggest possible trends in the expected direction. For response rate, a pattern showed that users might perform more similarly between MR and screen in high cognitive

load than low cognitive load as seen in table 2. Furthermore, regarding reaction time, a pattern is shown where participants might respond faster in MR at low load level, and slower in high load level. This can also be seen in figure 3. While it can be argued that the results indicate that the difference between modalities does depend on load level, the interaction effect was not significant ($p = .8$). These results are likely due to it being a big difference between people and a limited sample size. Even if the pattern is indicating an interaction, it is too small to be significant with 24 participants. Nevertheless, it is important to note that even though the interaction did not reach significance, the direction of the difference being reversed across load levels may suggest a trend toward an interaction that a larger sample size could have detected. However, since no performance metric showed statistical significance, no strong conclusions can be drawn. In addition, the success rate showed equal difference in median values between MR Low and Screen Low as between MR High and Screen High and for error rate, the median values were identical across all conditions not providing any support for hypothesis H3. To summarize, some results are showing a possible trend in line with H3, but it is not statistically significant and therefore H3 can not be confirmed.

5.4 Cognitive load as a mediating factor

In neither performance metric were a significant difference between cognitive load and presence observed. Even if the results indicate a trend towards a possible interaction, with reaction time and response rate showing patterns aligning with H3 and previous research suggesting a relationship between cognitive load and presence that influences performance [8], no difference were found and no strong conclusion can therefore be drawn. Furthermore, the crossover results in reaction time, showing a trend for better performance with MR under low load and better performance with the screen under high load, suggests a possible interaction and aligns with H2 and recent studies indicating that higher presence leads to better performance [7]. However, this was not observed for success rate and response rate, which instead contradicts H2. Regarding error rate, it was the only metric that showed a significant interaction between cognitive load and presence. However, follow-up tests showed no pairwise differences. Since the median scores were very low and nearly identical across all conditions, see table 2, there was little room for meaningful differences.

To summarize, no significant results were found for either performance metric. Even though the results suggest a trend towards H3, different performance metrics show different indications for H2. Together with no meaningful significance being found, no strong conclusion can be drawn regarding how cognitive load affects presence and performance in screen-based vs MR environments. This study shows that a possible interaction may occur, aligning with other studies indicating a possible link between presence and cognitive load [8, 19], although no firm conclusions can be drawn about how this interaction works. One reason for these results may be that presence did not achieve its full effect on learning as discussed in section 5.2, and that this also influenced how the results vary across different levels of cognitive load. This study's results suggest that further research would be beneficial, with a larger sample size to enable significant results that clearly demonstrate the type of interaction effect present.

5.5 Limitations

In order to create a scenario where no previous experience was necessary, to minimize learning biases and simulator sickness, as well as to make sure to the extent possible that no data were lost or compromised, the decision was made to not let participants steer the aircraft themselves. While this was

necessary for this study, it risked losing a possible important factor in learning transfer in virtual environments, that is, operating in the actual world. Another limitation was the length of the experiment, each participant only got to try each scenario for five minutes. This was to avoid the study taking too long and participants experiencing fatigue in the later scenarios since the conditions were tested consecutively. A potential improvement would be to run the tests over several days. The study was also conducted in a room without sound isolation, which created a risk of participants being disturbed by external noise that may have affected their sense of presence.

Another major limitation was the technology, which caused the flight to start at slightly different times for each participant. In other words, the time between pressing play and the full simulation loading varied between participants. Although the variation was not large, only a few milliseconds, it may have led some participants to see the numbers for the n-back task for a shorter time than others, since the number was pre-programmed to appear after a specific time in the game. While no major differences should have emerged from this, it remains a limitation. Furthermore, reaction time data from the n-back task was also collected, specifically measuring how quickly participants responded when detecting a target. However, this data could not be used in the analysis since the experimental conditions differed between participants. This data is therefore not included in the report.

5.6 Future work

Future work should include more participants in order to further investigate the relationship between cognitive load, presence and performance, and see whether any of the interaction effects would be proven significant. Another interesting direction would be to do a study with real pilots, thus enabling creating a more realistic scenario, allowing the participants to both steer the plane and create tasks requiring decision making, which requires previous knowledge. This way, the true effect of learning outcomes in regards to cognitive workload in flight simulation scenarios might be more visible. Furthermore, it would be beneficial to let the participants try the different conditions on different days, thus allowing for longer flying sessions and the effect of presence to be even more evident, as well as eliminating the risk of tiredness affecting the conditions tried later in the experiment.

6 CONCLUSION

This study supports the hypothesis that presence will be higher in both low and high cognitive load environments for MR-based simulation when compared to screen-based simulation. The study also found that, while results showed some indication towards an interaction effect between cognitive load and modality, no significant results were found. Together with inconsistent performance patterns, no firm conclusions can be drawn about how cognitive load influences performance across the two modalities. The study did not support that participants will perform better in MR compared to screen-based simulation when cognitive load is lower, since results varied across performance metrics and in some cases even contradicted it. Similarly, the study showed indication that there will be a smaller difference in performance, between MR vs Screen-based simulation, when the cognitive load is high compared to when the cognitive load is low, but this was not statistically supported. No significance was found for the interaction effect between modality and cognitive load. The results suggest a trend towards a possible interaction, and therefore further research with a larger sample size would be beneficial, as this would increase statistical power and allow for clearer conclusions about the interaction effect.

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Generative AI was used in this study to refine and improve the text.

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